Revise NCERT





Coordination Compounds

Grade 12 • Chemistry • Session 1





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Addition Compounds

They are formed by addition of 2 or more compounds

- Addition compounds are formed when stoichiometric amounts of two or more stable compounds join together.
- Examples:

Double salt

$$KCl + MgCl_2 + 6H_2O \rightarrow KCl.MgCl_2.6H_2O$$

(carnallite)

potash alum

$$K_2SO_4 + Al_2(SO_4)_3 + 24H_2O \rightarrow K_2SO_4.Al_2(SO_4)_3.24H_2O$$

(potassium alum)

$$CuSO_4 + 4NH_3 + H_2O \rightarrow CuSO_4.4NH_3.H_2O$$

(tetrammine copper(II) sulphate monohydrate)

$$Fe(CN)_2 + 4KCN \rightarrow Fe(CN)_2.\ 4KCN$$

(potassium ferrocyanide)

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Addition compounds

Double salts

$KCl \cdot MgCl_2 \cdot 6H_2O$

or

$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

→ form individual ions

→ lose identity in solutions.

Coordination compound

→ $CuSO_4 \cdot 4NH_3$

→ $Fe(CN)_2 \cdot 4KCN$

→ donot lose their identity

→ donot give tests of individual ions

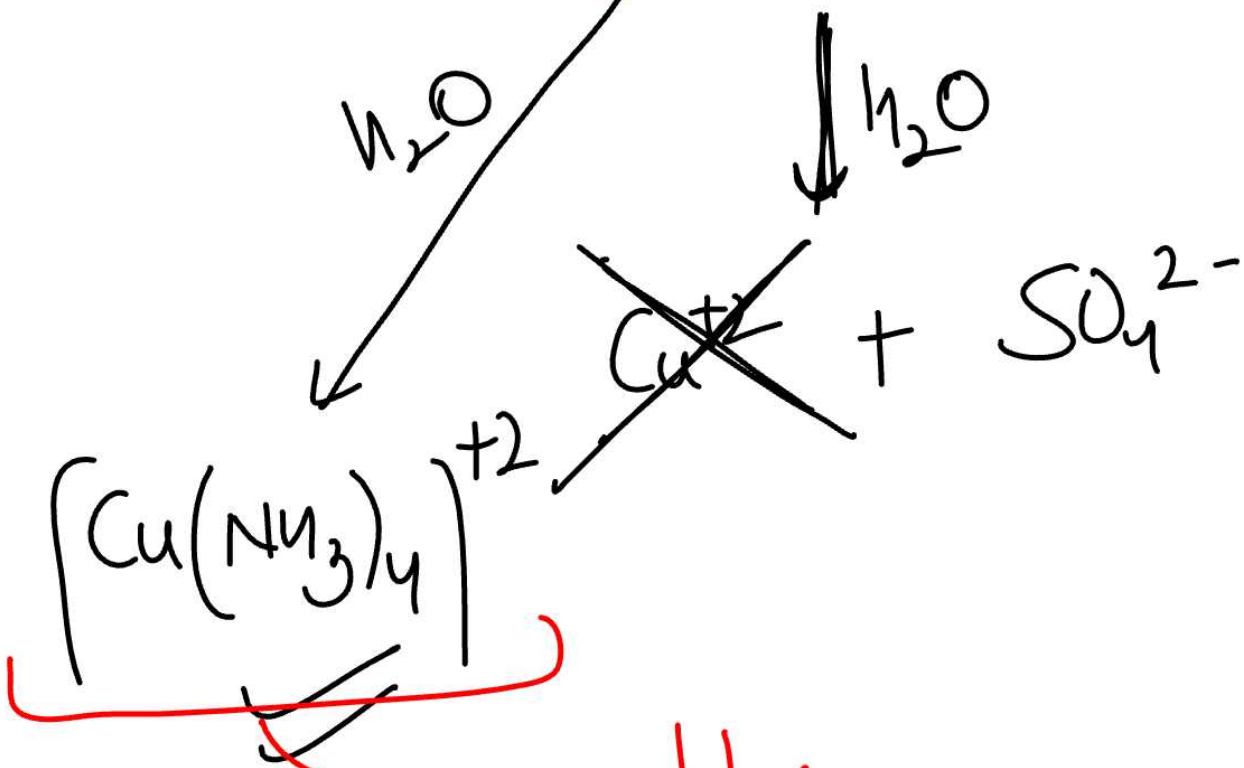
$KCl \cdot MgCl_2 \cdot 8H_2O \xrightarrow{H_2O} K^+ + Cl^- + Mg^{+2} + 2Cl^-$

$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O \xrightarrow{H_2O} 2K^+ + SO_4^{2-} + + 2Al^{+3} + 3SO_4^{2-}$

Double salts → They gives tests of individual ions

Coordination compounds →

$CuSO_4 \cdot 4NH_3$ exists as $[Cu(NH_3)_4]SO_4$



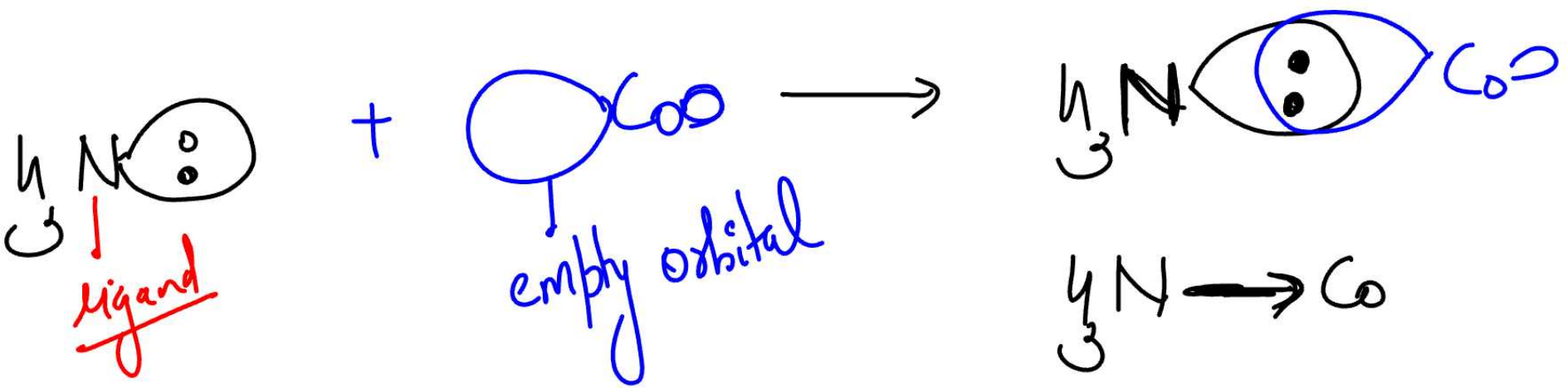
They donot give tests of individual ions.

Coordination compounds →

→ complex is enclosed in square bracket.

eg: 

Complex → contains coordinate bonds



Coordination compound → A coordination compound contains a metal atom/ion forming a no. of coordinate bonds with **ligands**.

They are also called complexes →

Types → (i) Neutral complexes → $[Ni(CO)_4]^0$, $[Fe(CO)_5]^0$

(ii) Cationic complexes → $[Co(NH_3)_6]^{+3}$, $[Mn(H_2O)_6]^{+2}$, $[Cr(NH_3)_6]^{+3}$ etc.

(iii) Anionic complexes → $[Fe(CN)_6]^{4-}$

or $[Fe(CN)_6]^{2-}$ etc

(iv) Homoleptic complexes → which contain only one type of ligand.

(V) Heteroleptic complexes → which contain two or more ligands.

eg: $[Fe(H_2O)_5(NO)]^{+3}$

$[Fe(CN)_3(NH_3)_3]$

Terms related to coordination compounds

1. Ligand → An e^- pair donor.

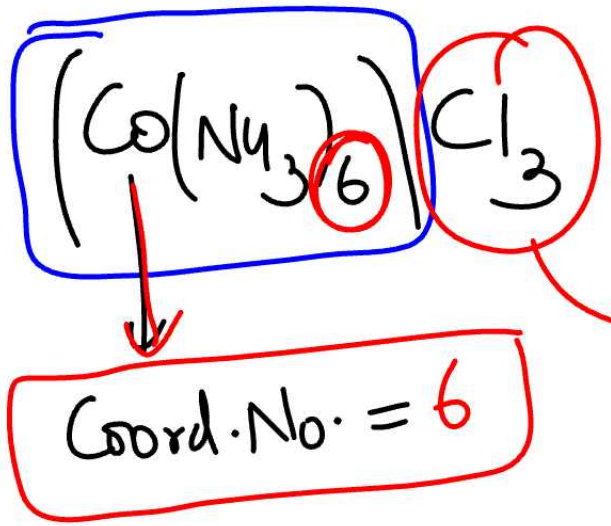
→ forms coordinate bonds with Central metal atom/ion by donating one or more l.p to empty orbital of Central metal atom/ion.

2. Coordination sphere → Central metal atom ion + ligands

$[Co(NH_3)_6]Cl_3$

Coordination sphere → non-ionisable.

Ionisation sphere → which can give free ions



Coordination no. → No. of coordinate bonds formed by central metal atom | ion.

Addition Compounds

Molecular or Addition Compounds → **Double salts**

- Stable in solid state only.
- Broken down into individual constituents when dissolved in water.
- Their solutions have the same properties as the mixture of individual compounds.



Carnallite ($KCl.MgCl_2.6H_2O$)

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Addition Compounds

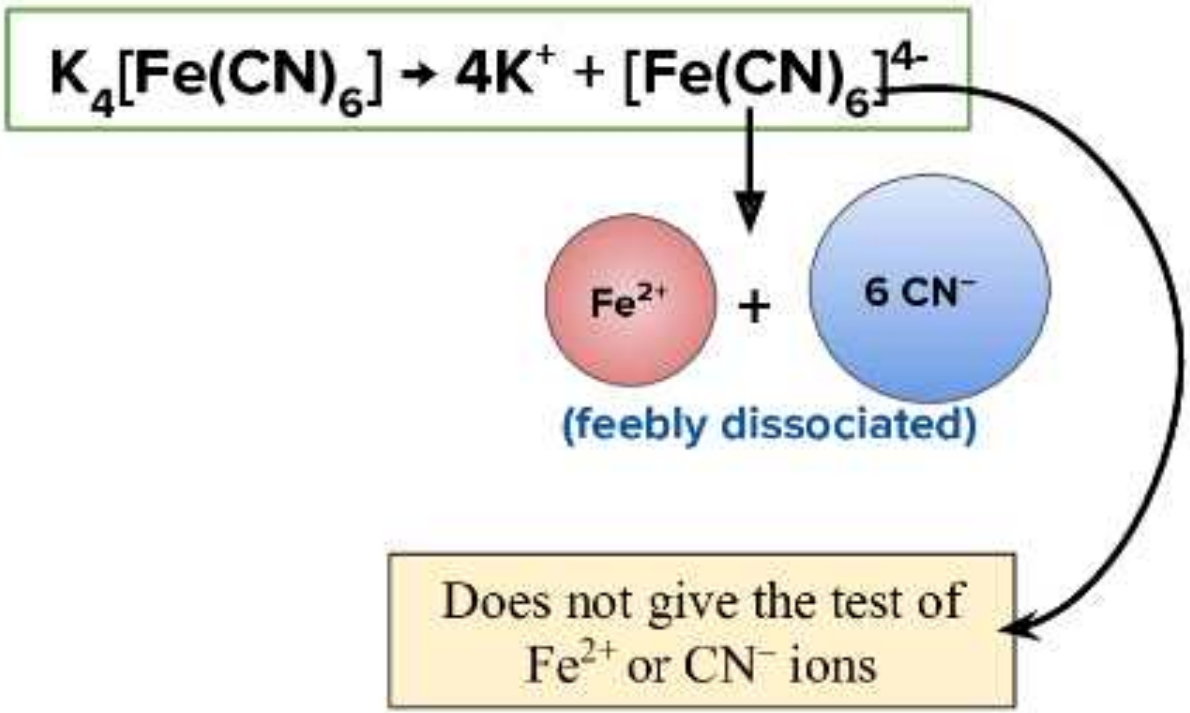
On the basis of stability, complex ions are divided into:



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Perfect Complexes

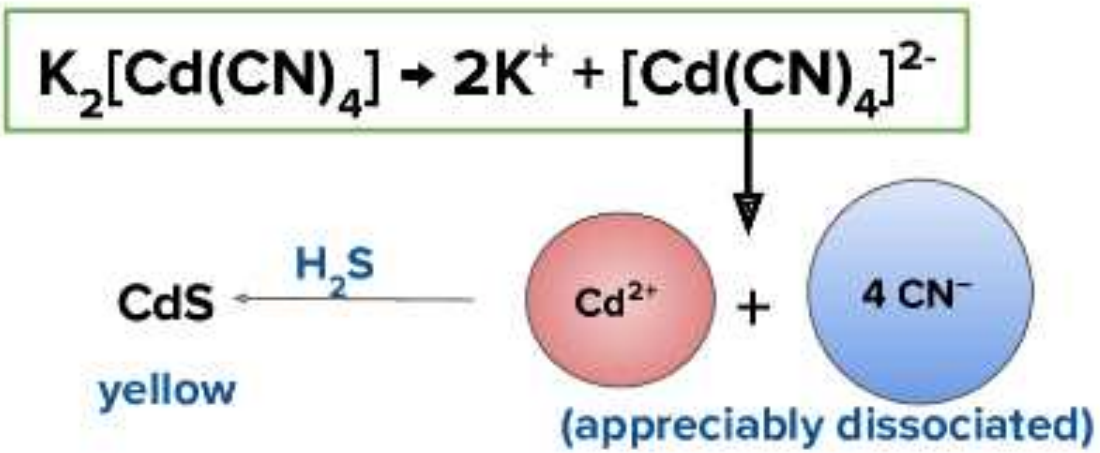
Complex ion is fairly stable and is either not dissociated or feebly dissociated in solution state.



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Imperfect Complexes

Complex ion is less stable and is reversibly dissociated to give enough simple ions and thus impart their tests.

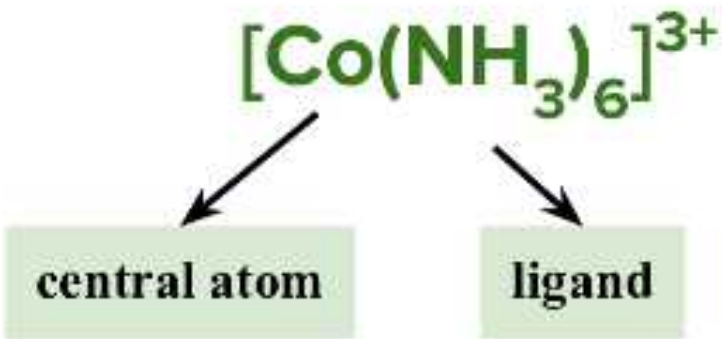


An imperfect complex may be too unstable to exist and may be completely dissociated in solution, it then becomes a double salts.

Coordination Entity/Coordination Sphere

A coordination entity constitutes a central metal atom or ion bonded to a fixed number of ions or molecules.

Example - $[Co(NH_3)_6]^{3+}$: cobalt ion is surrounded by six ammonia molecules.



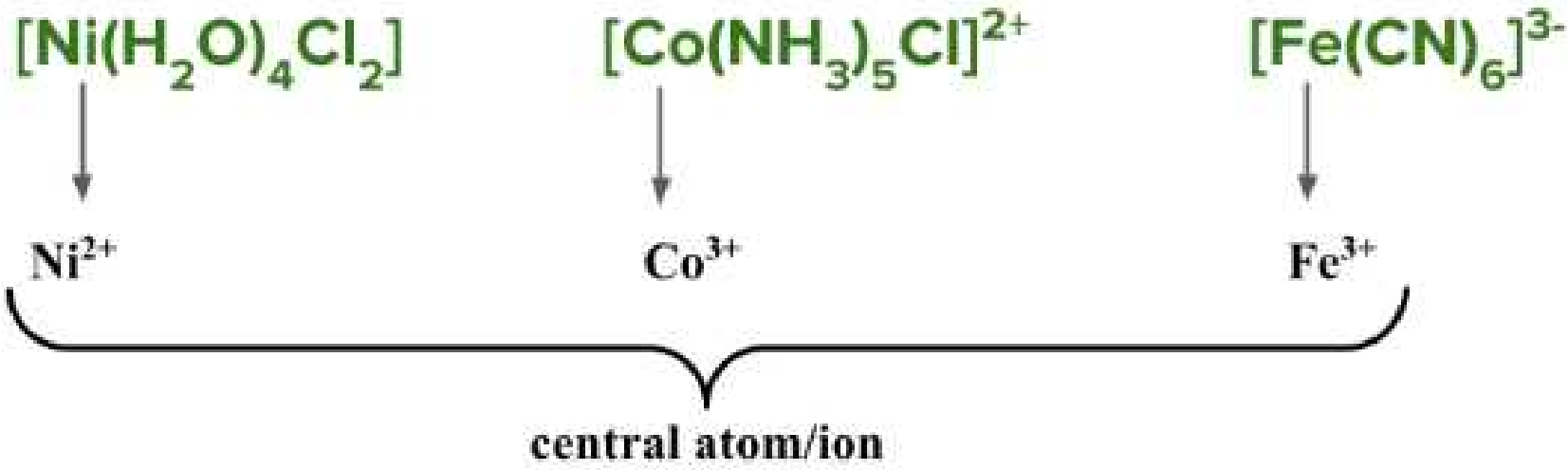
The central atom/ion and the ligands attached to it are enclosed in square bracket and is collectively called as **coordination sphere.**

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Central Atom/Ion

In a coordination entity - the **atom/ion** around which a fixed number of ligands are bound in a definite geometrical arrangement, is called the central atom or ion.

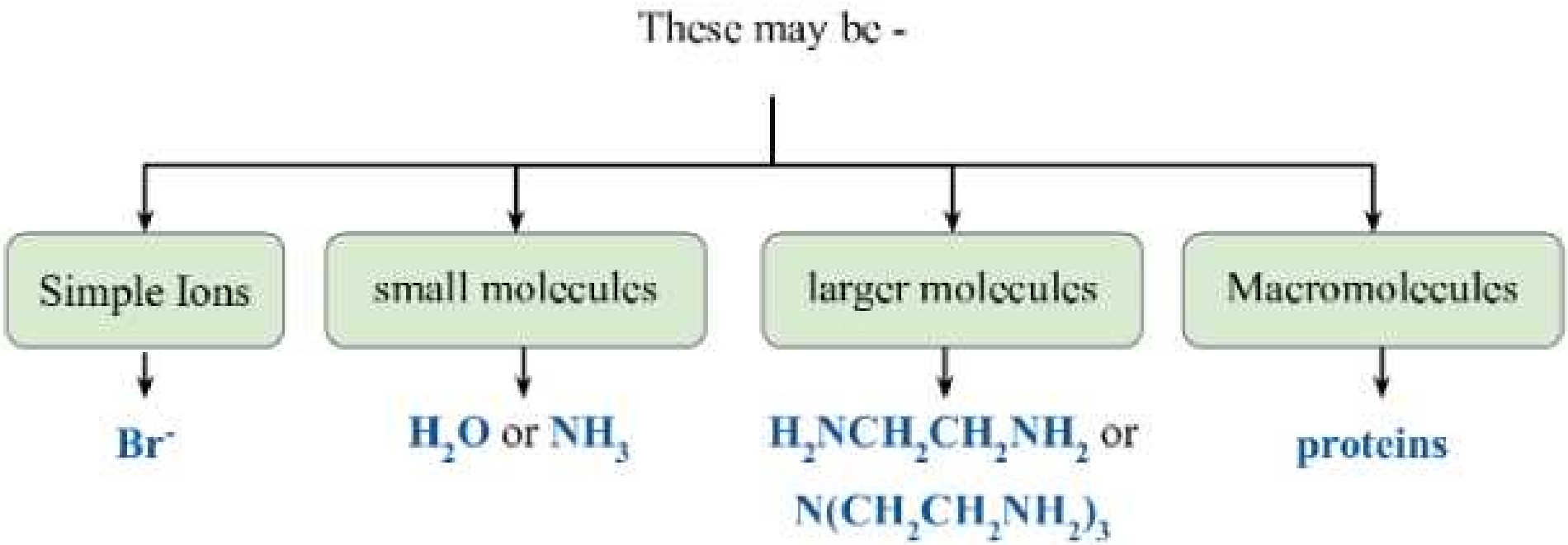
Example:



The central atoms/ions are also referred to as Lewis acids since they accept lone pairs from the ligands.

Ligands

The neutral molecules, anions or cations which are directly linked with central metal atom or ion in the coordination entity are called **ligands.**



The ligands are also referred to as Lewis bases since they donate lone pairs to the central metal atom/ion.

Counter Ions

A **counter ion** is the ion that accompanies an ionic species in order to maintain electrical neutrality. The ionisable groups written outside the bracket are called counter ions.

Example: $K_4[Fe(CN)_6]$

↓

K^+ → **Counter ion**

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Coordination Number

- The **coordination number** of the central atom/ion is determined by the no. of σ bonds between the ligands and the central atom/ion.

- The σ bonding e^- may be indicated by a pair of dots, preceding the donor atom in the ligand formula as in -

$[Co(:NH_3)_6]^{3+}$,	$[Fe(:CN)_6]^{3-}$,	$[Ni(:CO)_4]$,	$[Co(:Cl_4)]^{2-}$
6	6	4	4

Coordination number (C.N.) of 6, 4 and 2 are the most common C.N. and odd C.N. are generally not found (rare).

Coordination Number

Some common coordination number of important metals -

Metal	Coordination Number	Metal	Coordination Number
Cu^{+}	2, 4	Ni^{2+}	4, 6
Ag^{+}	2	Fe^{2+}	6
Au^{+}	2, 4	Fe^{3+}	6
Cu^{2+}	4, 6	Co^{2+}	4, 6
Pt^{2+}	4	Co^{3+}	6
Pd^{2+}	4	Pt^{4+}	6
Mg^{2+}	6	Pd^{4+}	6

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Oxidation number

- **Oxidation number of the central atom:** Charge on central atom if all the ligands are removed along with the e⁻ pairs that are shared with the central atom.

- Metal oxidation number is represented by a **Roman numeral** in parentheses.

Example:

Oxidation number of iron in $[Fe(CN)_6]^{3-}$ is **+3** and it is written as **Fe(III)**

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Quiz.

Chat Off

An aqueous solution of potash alum($K_2SO_4.Al_2(SO_4)_3.24H_2O$) gives

A. 2 types of ions

B. Only 1 type of ion

C. 4 types of ions

D. 3 types of ions

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Quiz.

Chat On

An aqueous solution of potash alum($K_2SO_4.Al_2(SO_4)_3.24H_2O$) gives

A. 2 types of ions

B. Only 1 type of ion

C. 4 types of ions

D. 3 types of ions

Solution

$K_2SO_4.Al_2(SO_4)_3.24H_2O$ will give K^+, Al^{3+} and SO_4^{2} ions.

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Quiz.

Chat Off

Which of the following complexes is homoleptic?

A $[Co(NH_3)_4Cl_2]^+$

B $[Ni(CN)_4]^{2-}$

C $[Ni(NH_3)_4Cl_2]$

D $[Fe(NH_3)_4Cl_2]^+$

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Quiz.

Chat On

Which of the following complexes is homoleptic?

A $[Co(NH_3)_4Cl_2]^+$

B $[Ni(CN)_4]^{2-}$

C $[Ni(NH_3)_4Cl_2]$

D $[Fe(NH_3)_4Cl_2]^+$

Solution

Homoleptic complexes have only one type of ligand.

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Quiz.

Chat Off

What is the oxidation number of central metal ion in $[Ni(NH_3)_5Br]^+$?

A. +1

B. +2

C. +3

D. +4

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Quiz.

Chat On

What is the oxidation number of central metal ion in $[Ni(NH_3)_5Br]^+$?

A. +1

B. +2

C. +3

D. +4

Solution

$x + 5(0) + 1(-1) = +1$

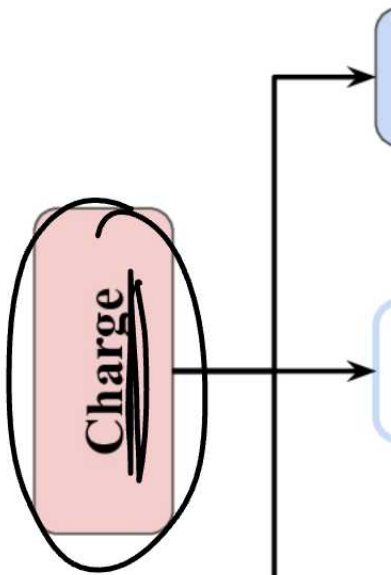
$x - 1 = 1$

$x = +2$

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Classification of Ligands

NH_3 → ammine



- **Neutral**
- Positive
- Negative

Examples:

CO (carbonyl)

$H_2N\,CH_2.CH_2NH_2$ (ethylenediamine) (en)

H_2O (aqua)

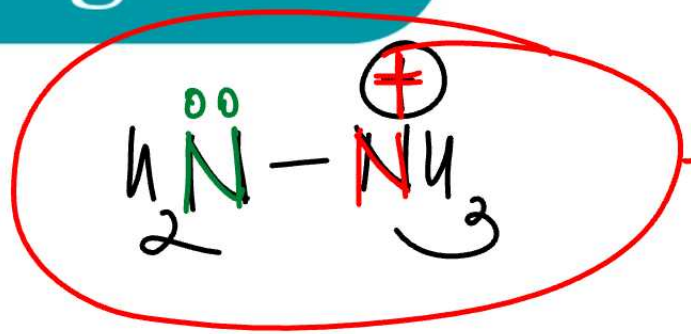
NH_3 (ammine)

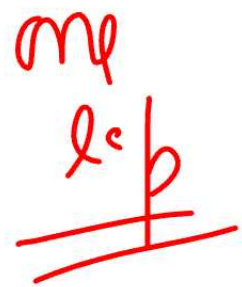
$H_2N-CH_2CH_2-NH_2$

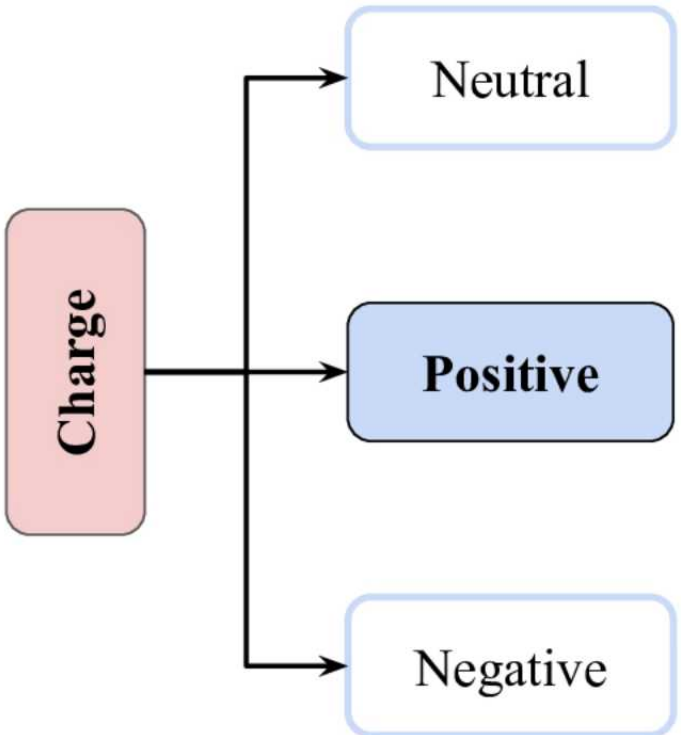
ethylenediamine

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Classification of Ligands







Examples:

$NH_2NH_3^+$ (hydrazinium)

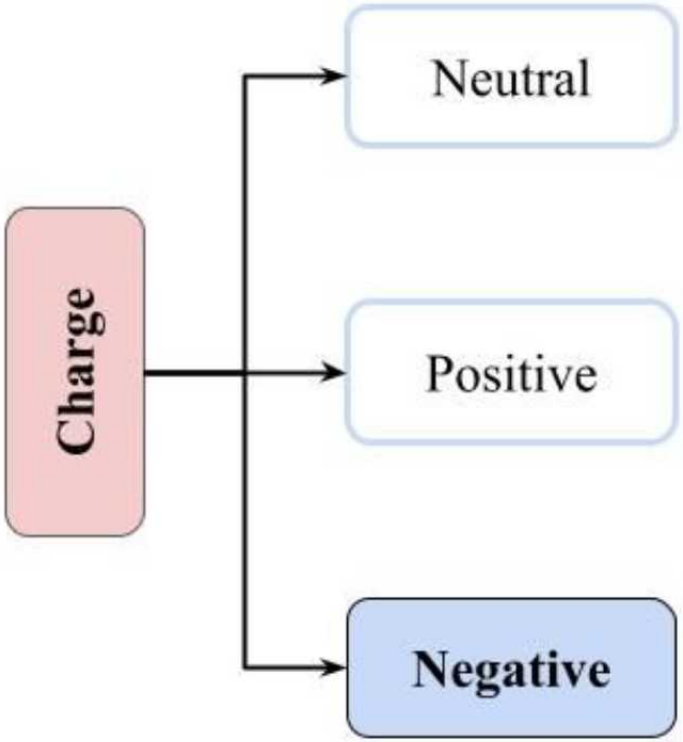
NO_2^+ (nitronium)

NO^+ (nitrosonium).



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Classification of Ligands



for -ve ligands

replace e ⟶ o

Cl^- – Chloride

$\bar{C}N$ – Cyanido

F^- – Fluorido

Examples:

Cl^- (chlorido), CN^- (cyanido), Br^- (bromido)

OH^- (hydroxo), CO_3^{-2} (carbonato), SCN^- (thiocyanato)

SO_4^{2-} (sulphato), $C_2O_4^{2-}$ (oxalato), NO_2^- (nitrito)

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F^- Cl^- Br^- I^-

halo

CN^-

pseudo halids

F^- — Fluoro Cl^- → Chloro

Br^- — Bromo I^- — Iodo

$\bar{C}N$ — cyano

Classification of Ligands

Denticity → No. of e⁻ pairs donated by 1 molecule of ligands

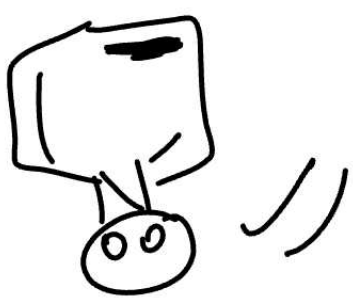
- Denticity
  - Monodentate
  - Bidentate
  - Tridentate
  - Tetradentate
  - Pentadentate
  - Hexadentate

They have one donor atom, i.e they provide only one e^- pair to central metal atom or ion.

Examples:

F^-, Cl^-, Br^-, H_2O, NH_3, CN^-

NO_2^-, OH^-, CO, Py, H^-



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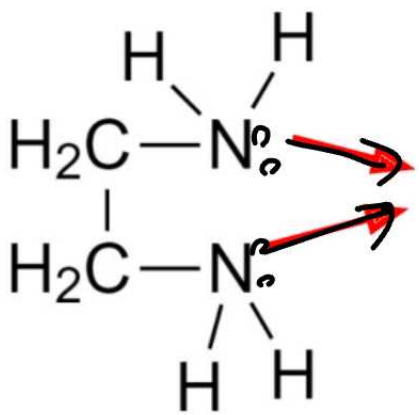
Classification of Ligands

- Denticity
  - Monodentate
  - **Bidentate**
  - Tridentate
  - Tetradentate
  - Pentadentate
  - Hexadentate

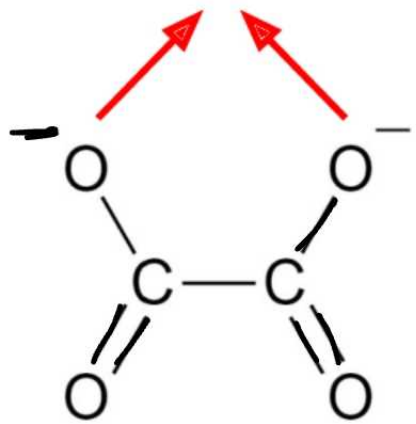
donate 2 l.p from 2 donor atom to same metal ion.

They have two donor atom (each) and have the ability to link with central metal ion at two positions.

Examples:



Ethylenediamine(en)



Oxalate(ox)

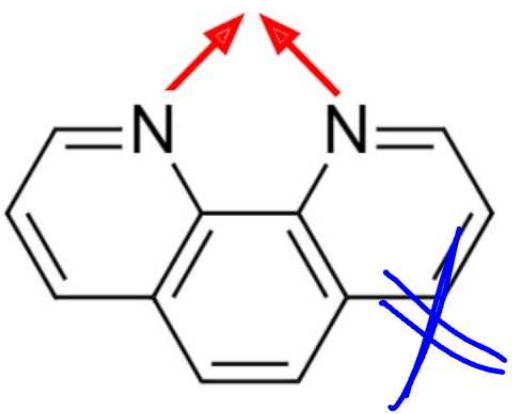
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Classification of Ligands

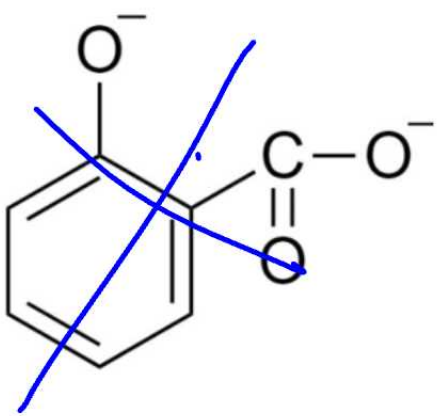
- **Denticity**
  - Monodentate
  - **Bidentate**
  - Tridentate
  - Tetradentate
  - Pentadentate
  - Hexadentate

They have two donor atom and have the ability to link with central metal ion at two positions.

Examples:



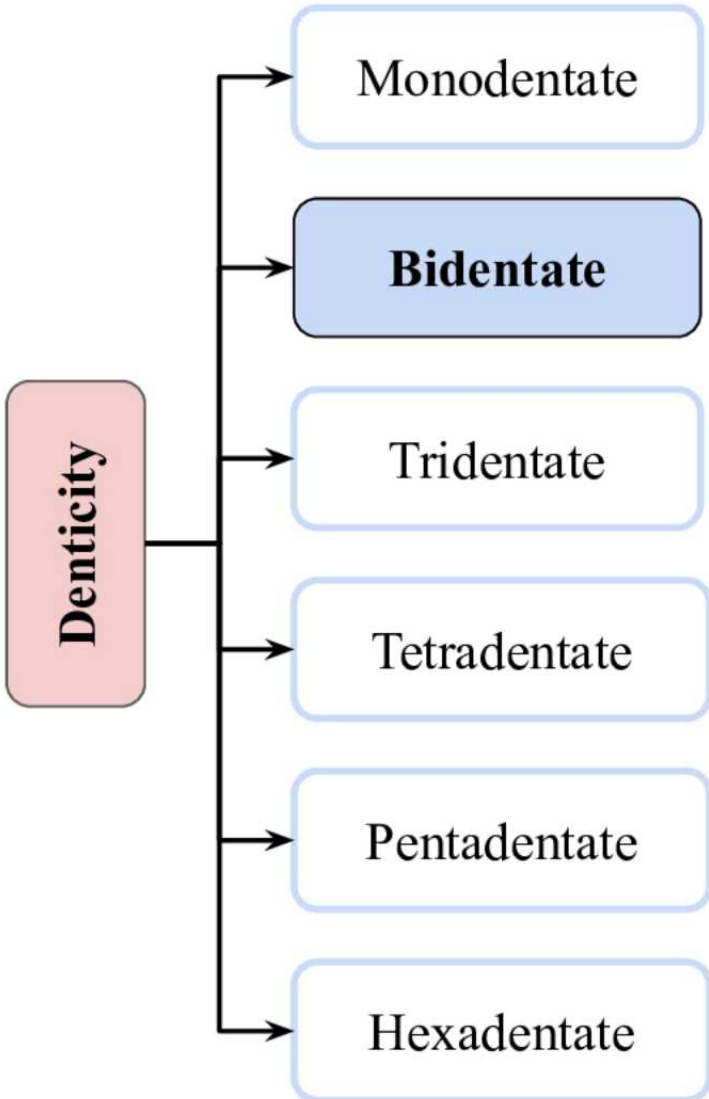
1,10-Phenanthroline (o-phen)



Salicylato (sac)

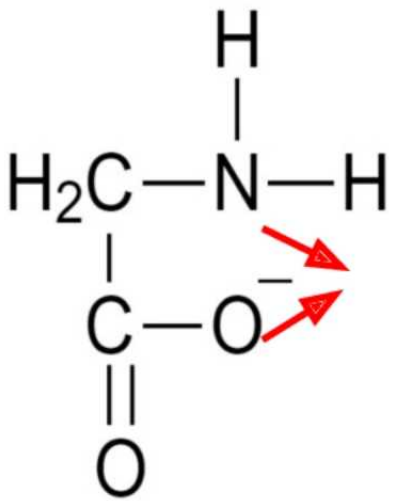
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Classification of Ligands

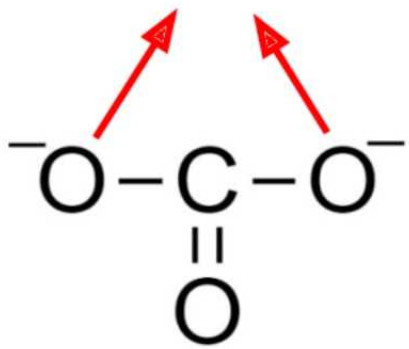


They have two donor atom and have the ability to link with central metal ion at two positions.

Examples:



Glycinato (Gly)

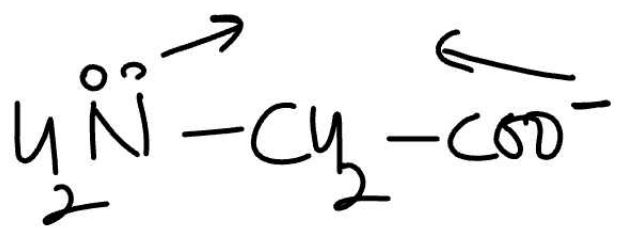


Carbonato

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Bidentate ligands →

en , ox , gly (glycinate)



Quiz.



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Quiz.

Chat Off

All ligands are

A Lewis acid

B Lewis base

C Neutral

D Reducing agent

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Quiz.

Chat On

All ligands are

A Lewis acid

B Lewis base

C Neutral

D Reducing agent

Solution

Since ligands are electron pair donors, they are all lewis bases.

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Quiz.

Chat Off

Bidentate ligand is

A pyridine

B acetylacetonate

C SCN^-

D Cl^-

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Quiz.

Chat On

Bidentate ligand is

A pyridine

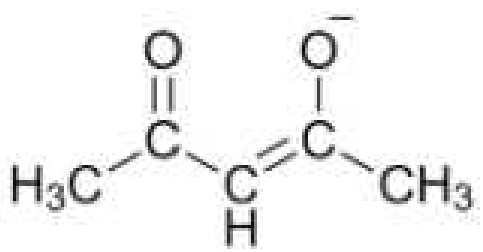
B acetylacetonate

C SCN^-

D Cl^-

Solution

Acetylacetonate is a bidentate ligand.



Acetylacetonato (acac)

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Quiz.

Chat Off

Ethylenediamine (en) is a

A Neutral ligand

B Monodentate ligand

C Bidentate ligand

D Both (A) and (C)

A, B, C

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Quiz.

Chat On

Ethylenediamine (en) is a

A Neutral ligand

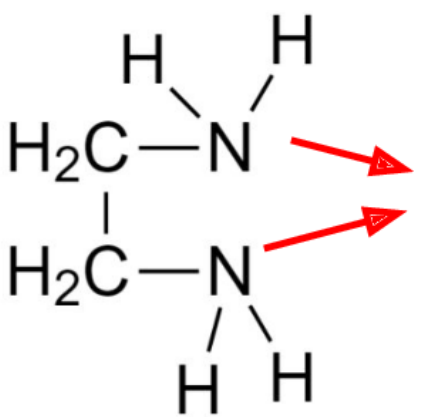
B Monodentate ligand

C Bidentate ligand

D Both (A) and (C)

Solution

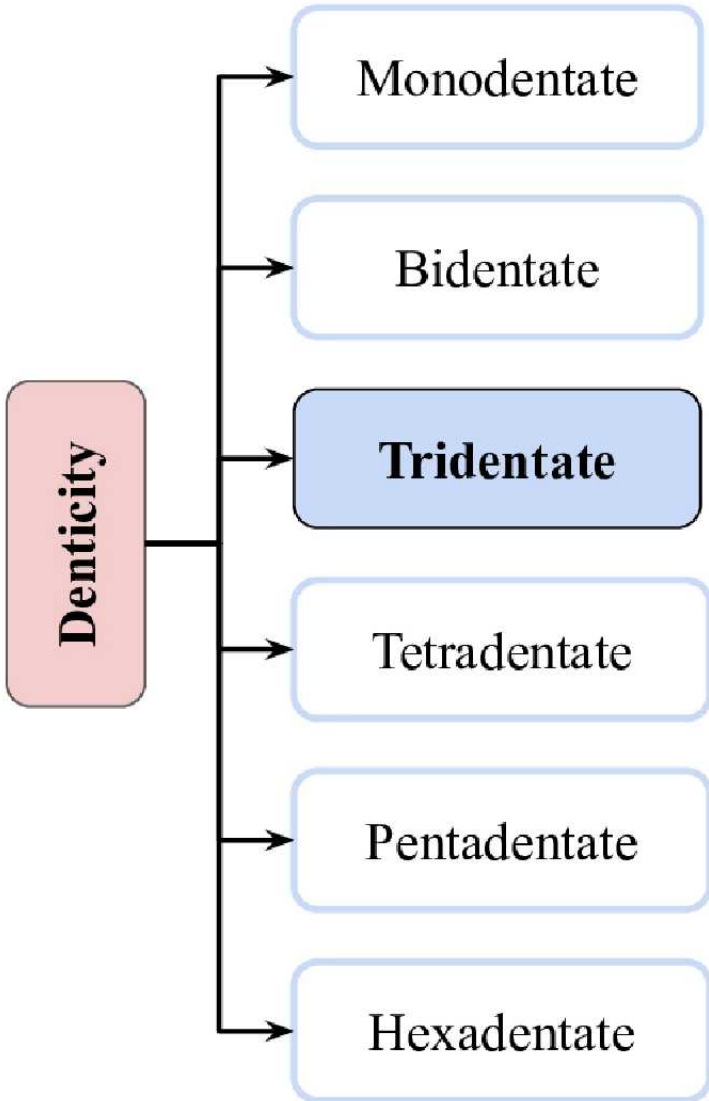
It is both bidentate and neutral.



Ethylenediamine(en)

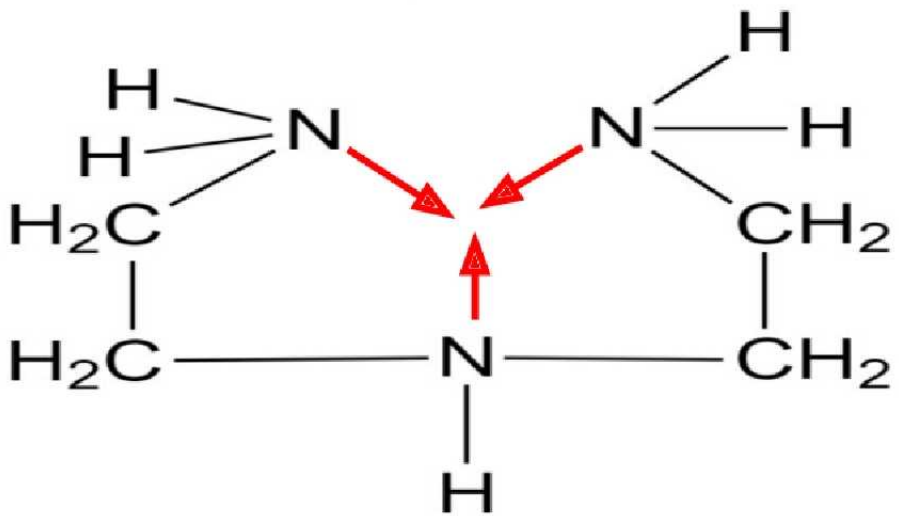
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Classification of Ligands



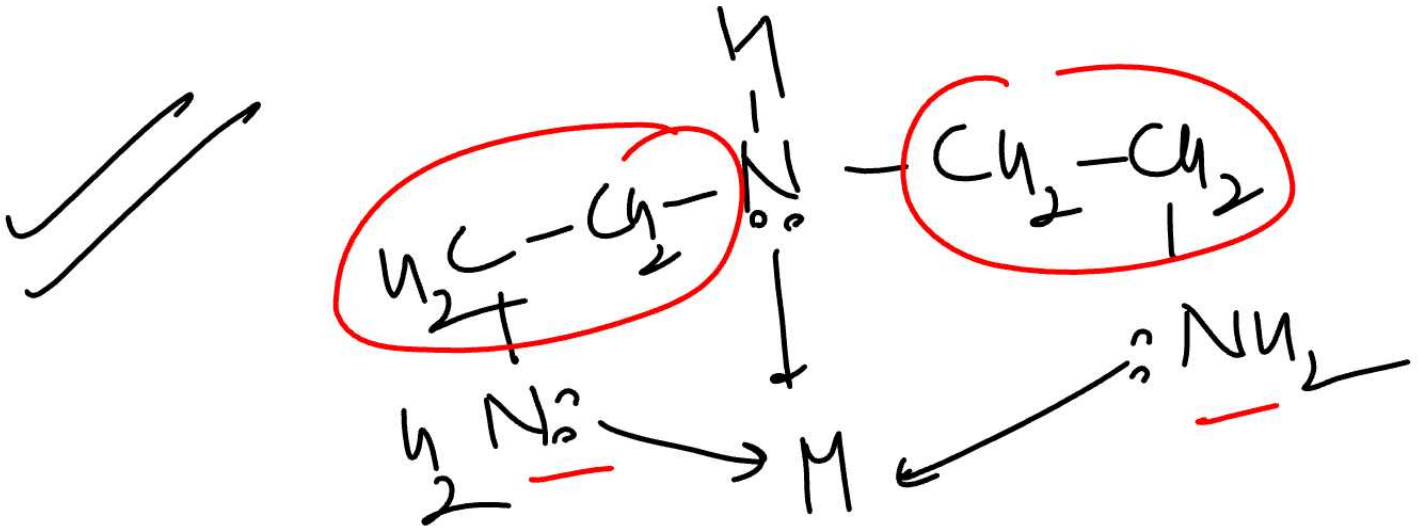
The ligands having three donor atoms are called tridentate ligands.

Examples:



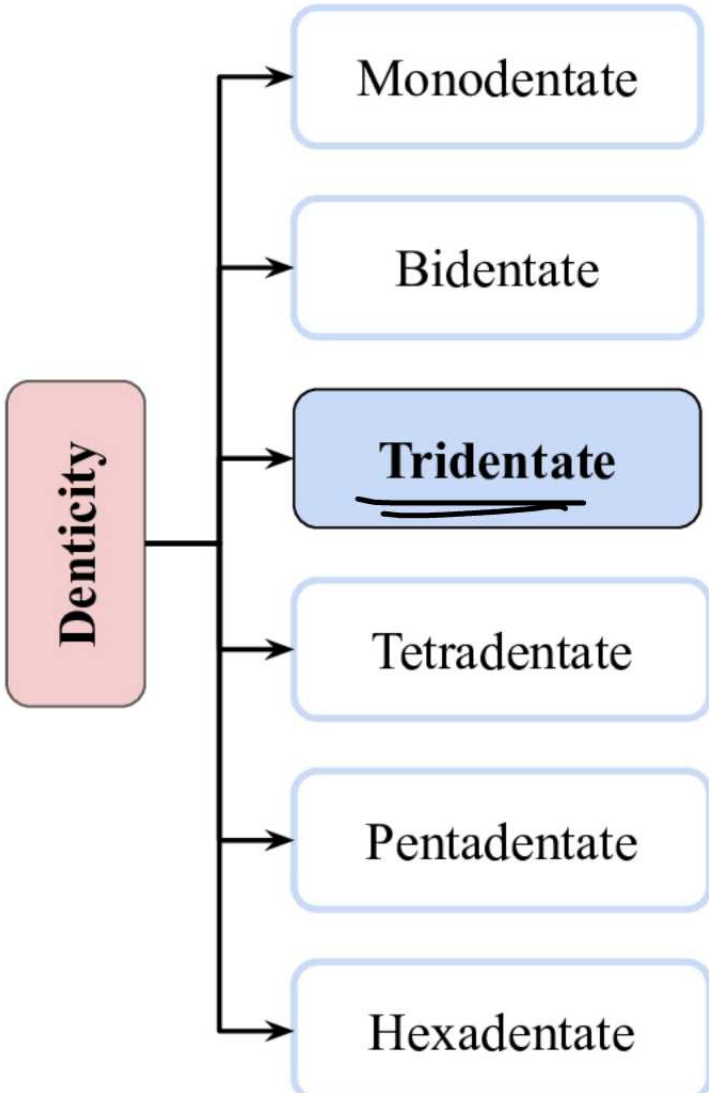
Diethylene triamine (Dien)

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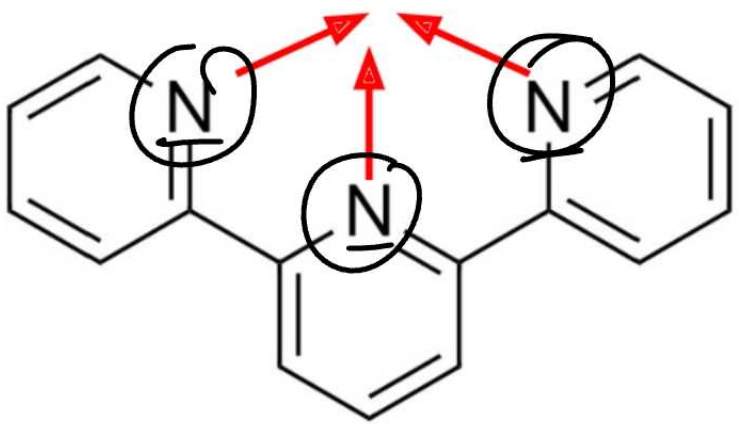
Diethylenetriamine (dien)

Classification of Ligands



The ligands having three donor atoms are called tridentate ligands.

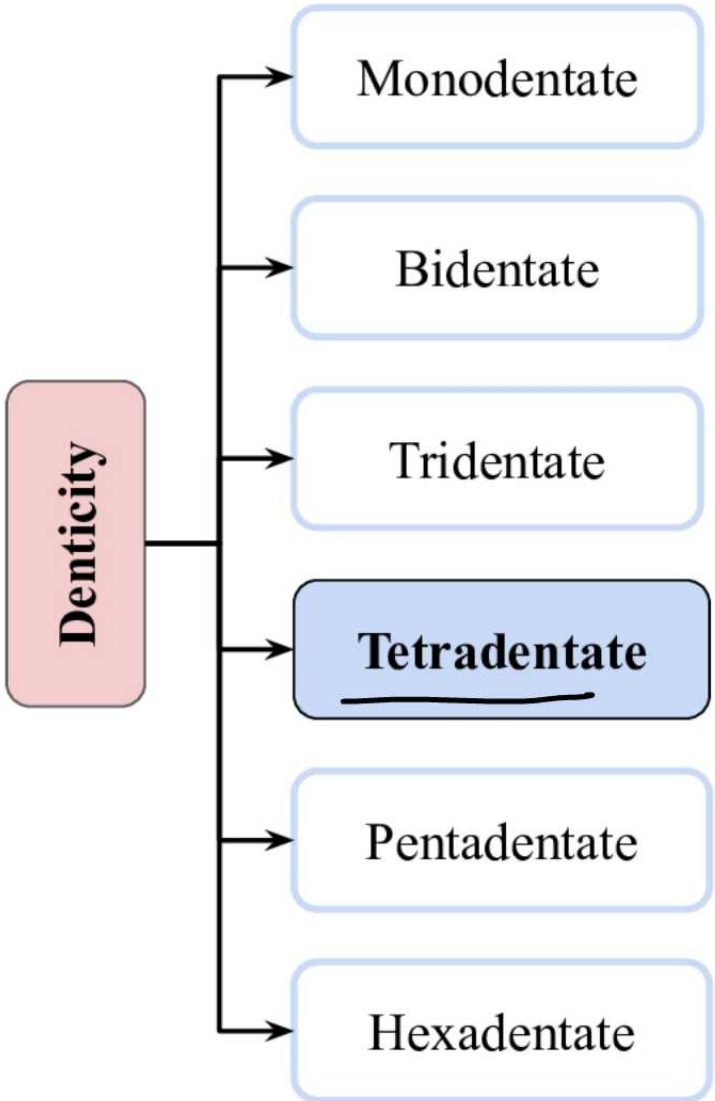
Examples:



2,2',2"-Terpyridine (terpy)

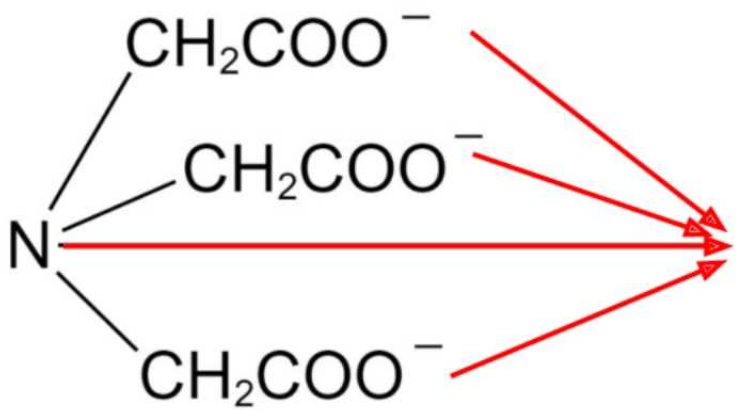
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Classification of Ligands



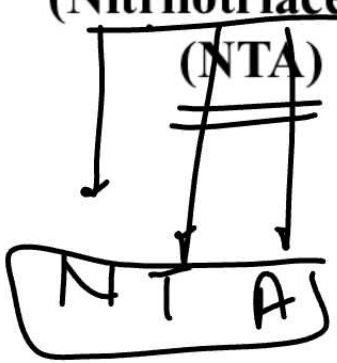
The ligands having four donor atoms are called tetradentate ligands.

Examples:

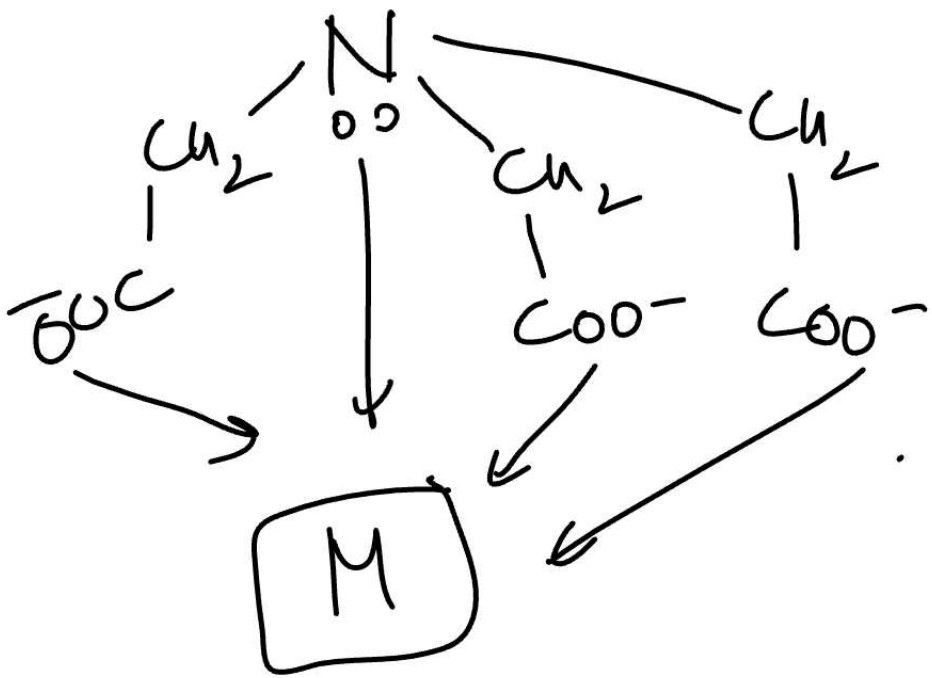


(Nitrilotriacetato)

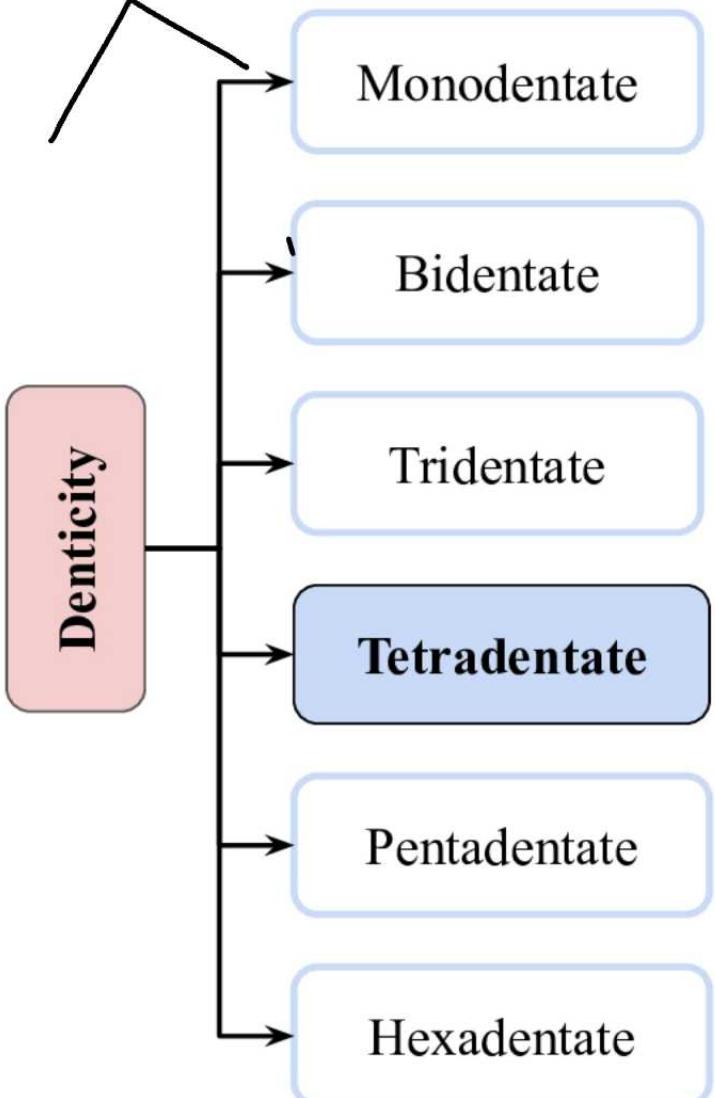
(NTA)



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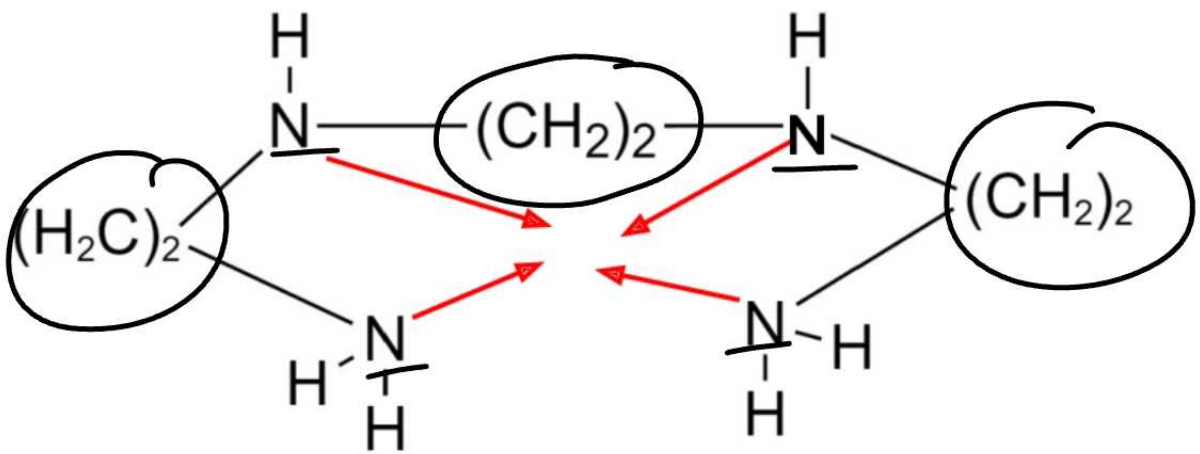

N
CH2
CH2
CH2
OOC
COO-
COO-
M

Classification of Ligands



The ligands having four donor atoms are called tetradentate ligands.

Examples:



Triethylene tetramine (Trien)

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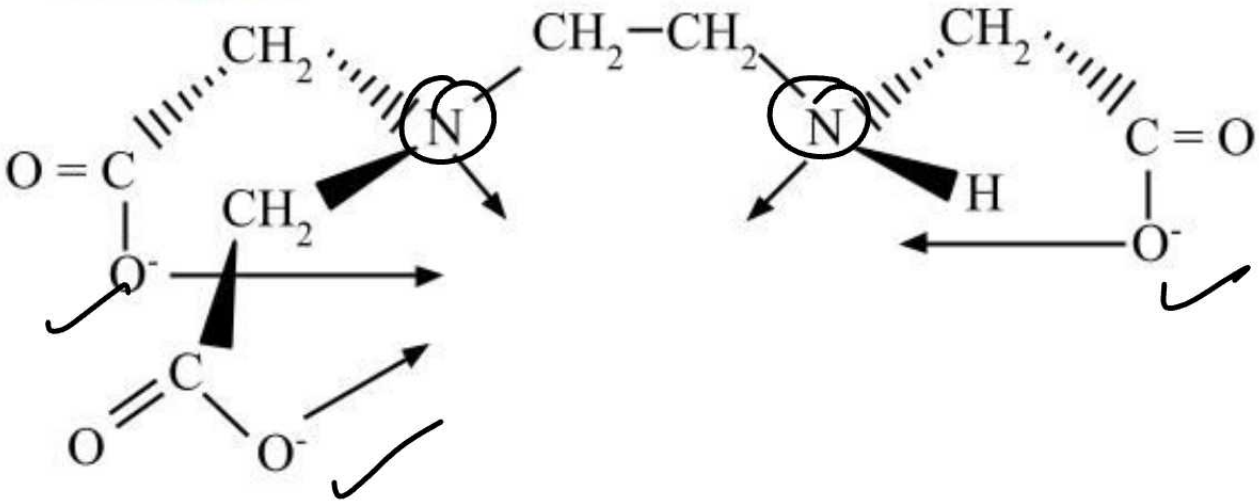
Classification of Ligands

Denticity

- Monodentate
- Bidentate
- Tridentate
- Tetradentate
- **Pentadentate**
- Hexadentate

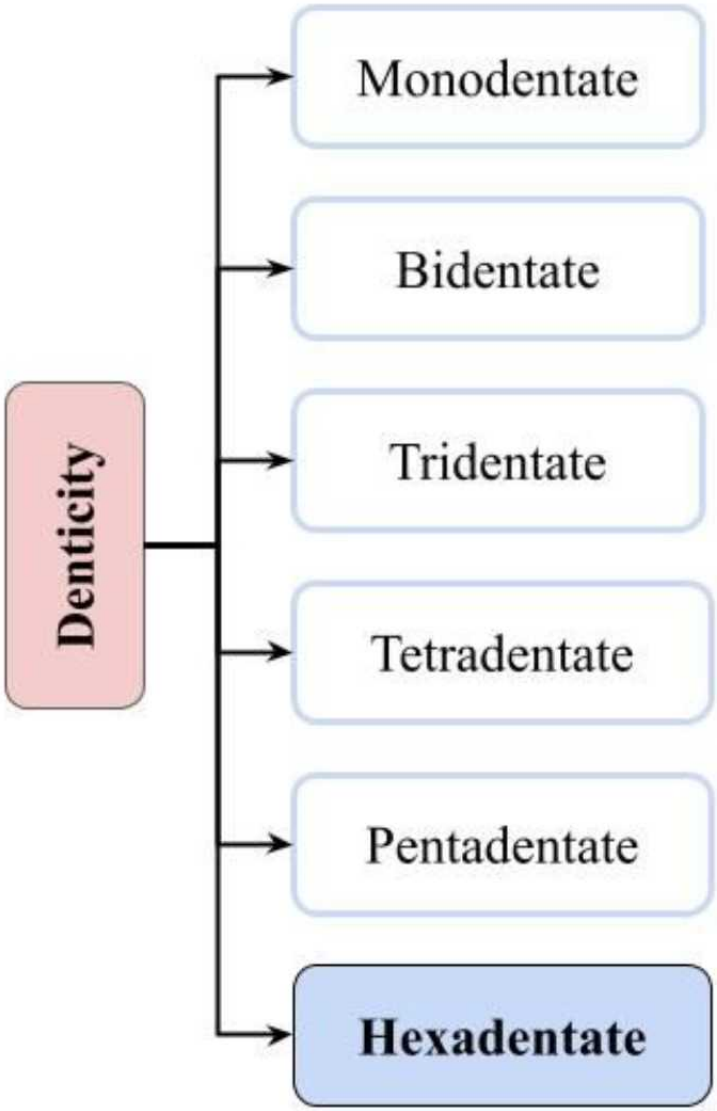
The ligands having five donor atoms are called pentadentate ligands.

Examples:



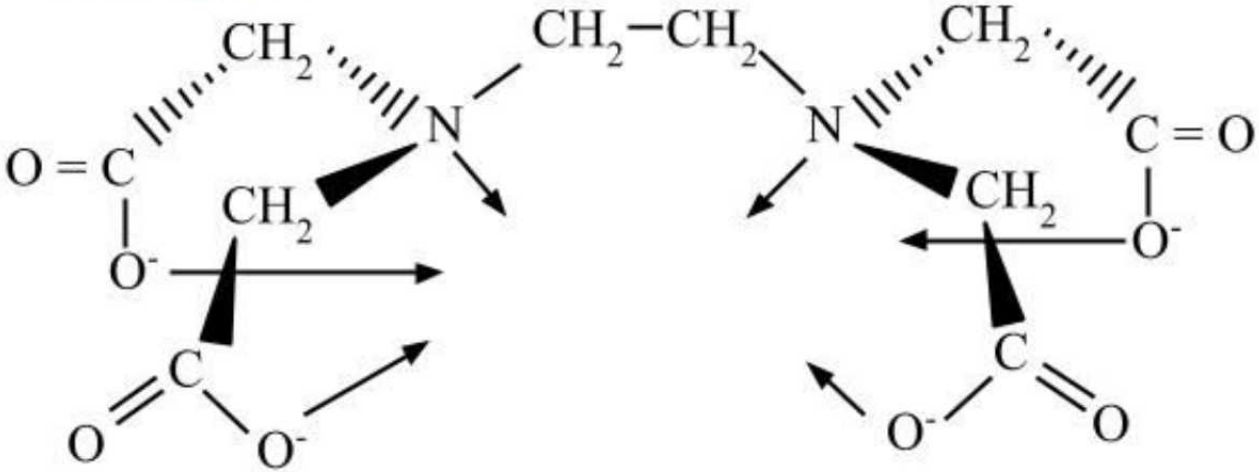
Ethylenediaminetriacetate ion $(EDTA)^{3-}$

Classification of Ligands



The ligands having six donor atoms are called hexadentate ligands.

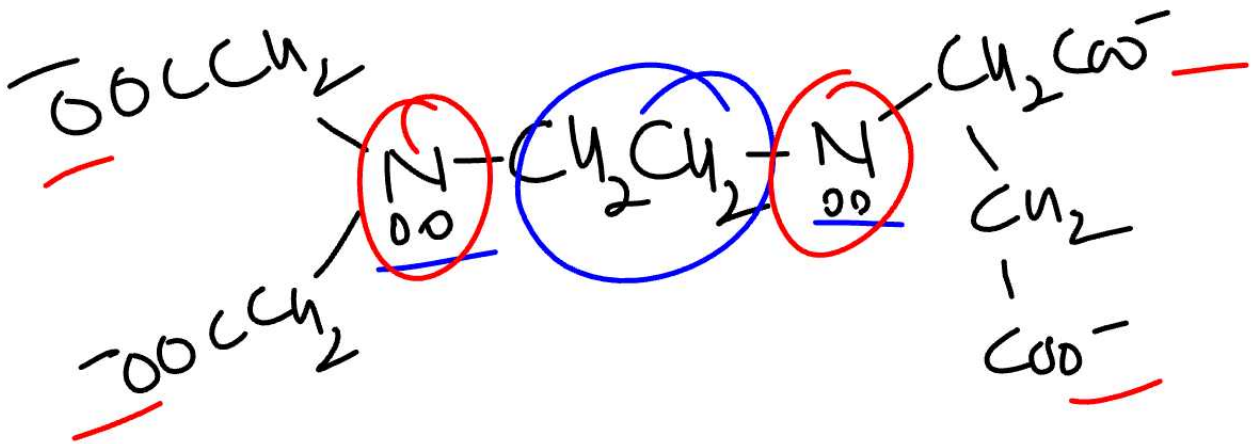
Examples:



Ethylenediaminetetraacetate ion $(EDTA)^{4-}$

$(EDTA)^{4-}$

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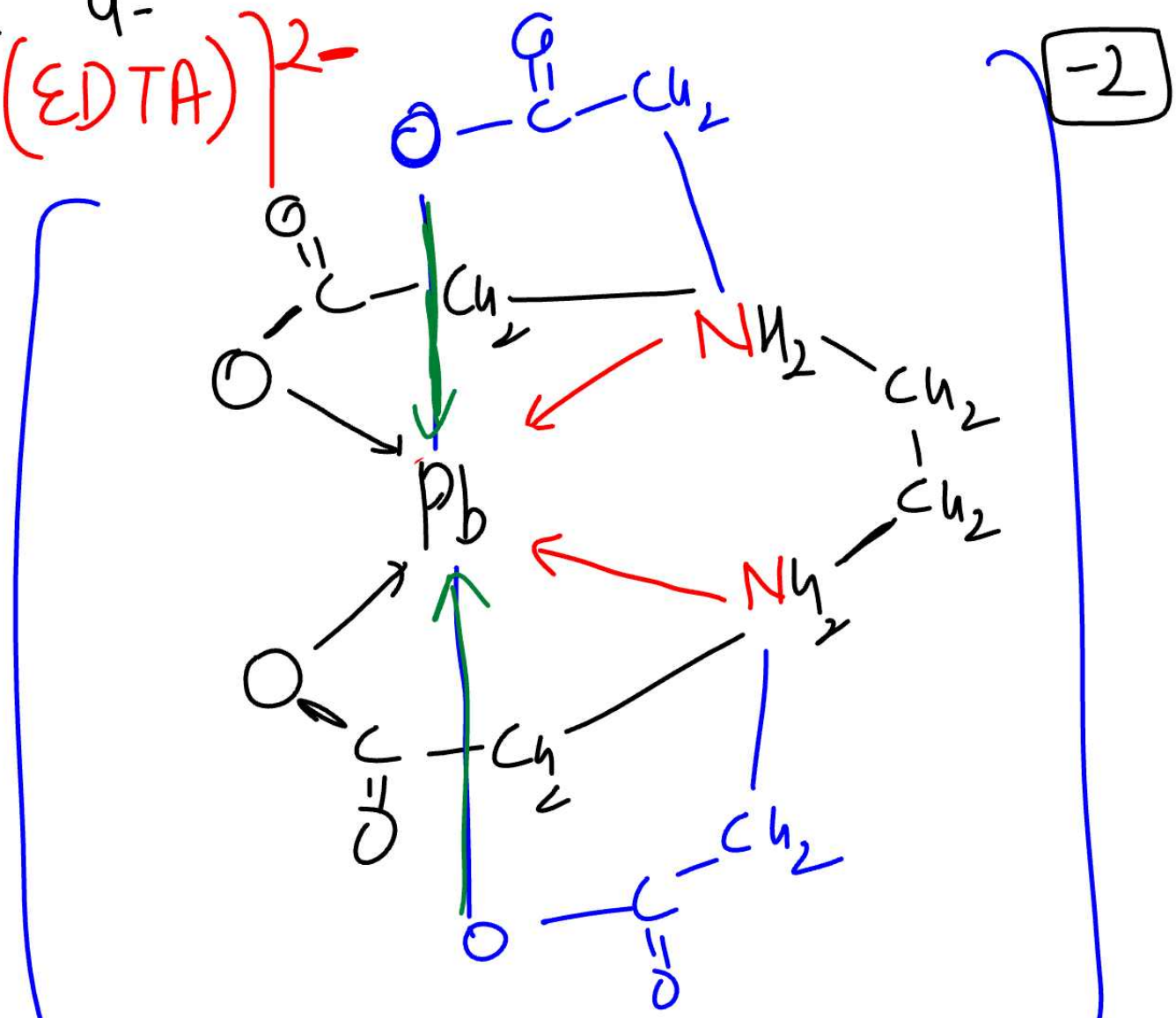
Ethylenediaminetetraacetato

$(EDTA)^{4-}$

Use → (i) It is used to remove hardness of water bec it forms stable complexes with Mg^{+2} & Ca^{+2}

(ii) It is used to remove Pb^{+2}.

$[Pb^{+2}(EDTA)^{4-}]^{2-}$



Coord. No. of Pb = 6

$+2 + (-4) = -2$

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Interactivity Pod

$[EDTA]^{3-}$ is a ____ ligand.

A pentadentate

B hexadentate

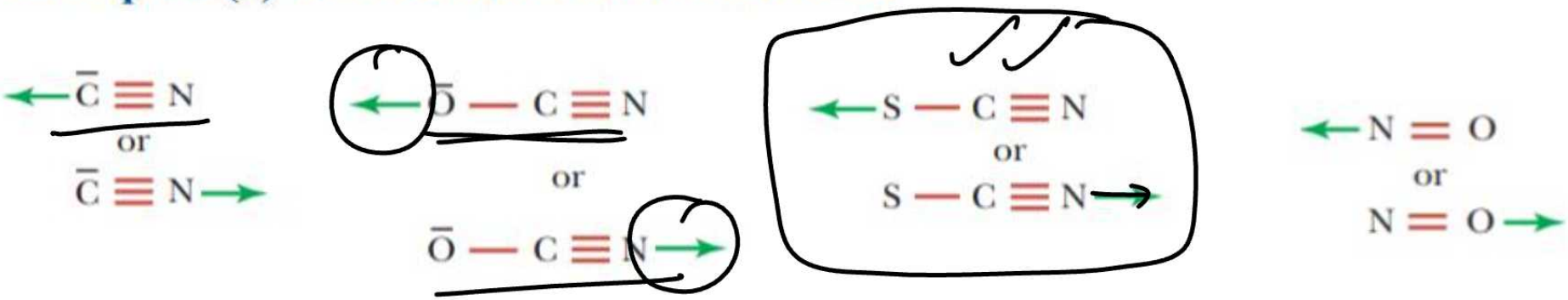
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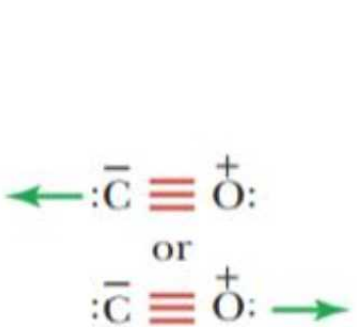
Classification of Ligands

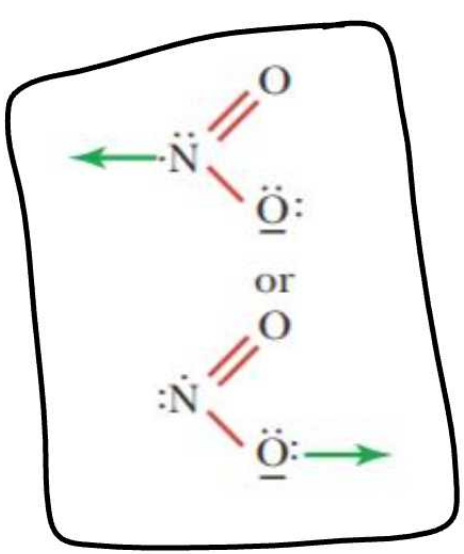
Ambidentate ligand

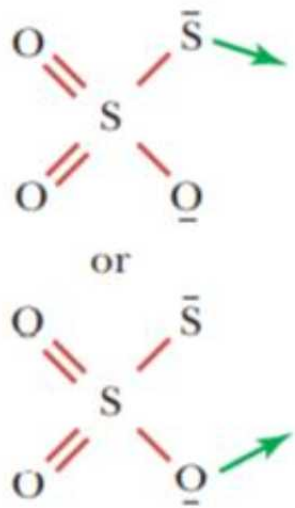
A ligand that may have more than one kind of donor sites but at a time only one kind of donor site is utilized for donation is called as ambidentate ligand.

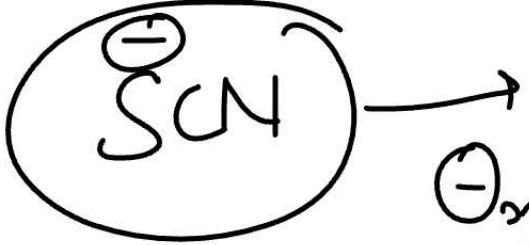
Examples: (a) Monodentate and ambidentate:



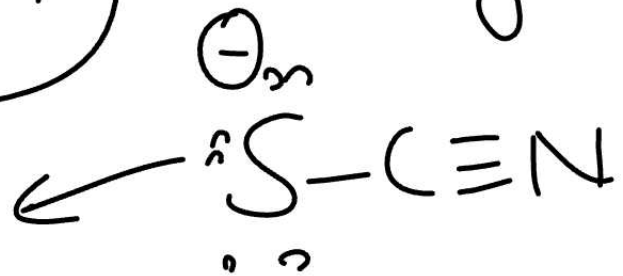




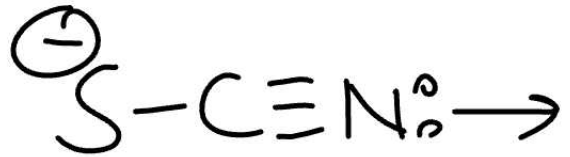




→ thiocyanate



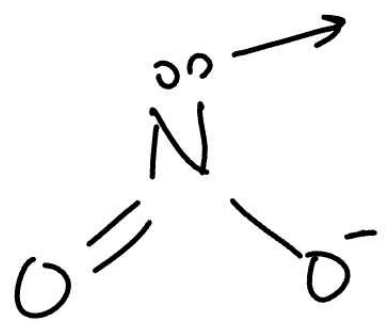
thiocyanato-S
(SCN)



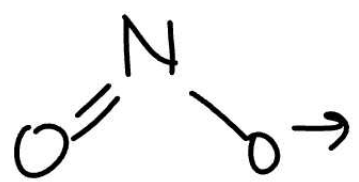
→ Isothiocyanato

or thiocyanato-N
(NCS)

(ii) NO_2^-



nitrito-N
(NO_2)



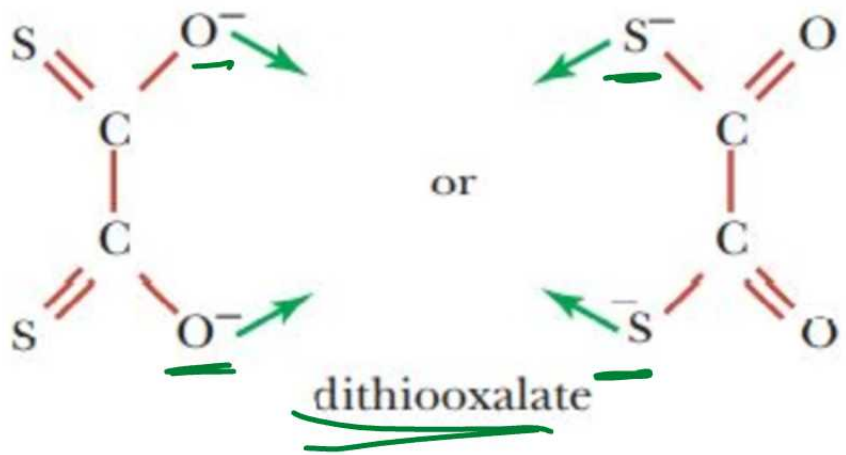
nitrito-O
(ONO)

Classification of Ligands

Ambidentate ligand

A ligand that may have more than one kind of donor sites but at a time only one kind of donor site is utilized for donation is called as ambidentate ligand.

Examples: (b) bidentate and ambidentate:



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Interactivity Pod

Which of these ligands is both bidentate and ambidentate?

A $S_2C_2O_2^{2-}$

B $S_2O_3^{2-}$

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Interactivity Pod

Which of these ligands is both bidentate and ambidentate?

A $S_2C_2O_2^{2-}$

B $S_2O_3^{2-}$

$S_2C_2O_2^{2-}$: bidentate and ambidentate

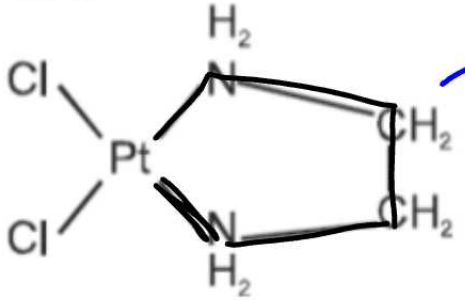
$S_2O_3^{2-}$: monodentate and ambidentate

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Classification of Ligands

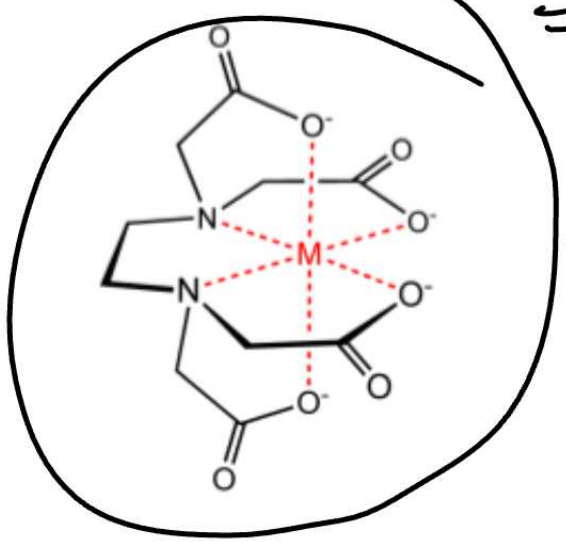
Chelating ligand → Polydentate ligand

- A ligand that can form a ring structure with the central atom is called a chelating ligand.
- All polydentate ligands are the example of chelating ligands.



→ Chelates

- Some ligands can form more than one ring; ex. $(EDTA)^{4-}$



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Classification of Ligands

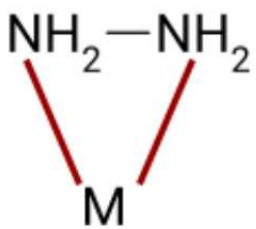
Chelating ligand

Chelation: The complex formed is referred to as a chelate complex and the process of chelate formation is called chelation.

- Chelate rings may have any number of atoms; the most common contain five or six atoms, including the metal ion.
- Smaller rings have smaller angles and distances that **lead to strain**.
- Larger rings frequently result in crowding both within the ring and between adjoining ligands.

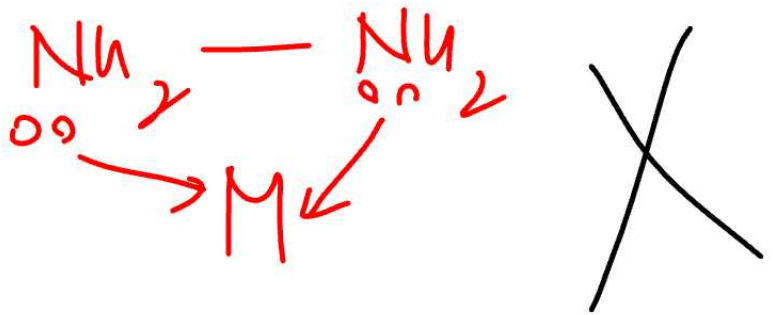
ring having > 6

NOTE: NH_2NH_2 and $NH_2(CH_2CH_2)_3NH_2$ cannot act as chelating ligands due to the formation of three membered ring and locked structure, respectively.



Three member highly unstable strained ring.

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NH2
NH2
M

Classification of Ligands

Chelating ligand

- Chelated complexes are **more stable** than similar complexes with monodentate ligands as dissociation of the complex involves breaking two bonds rather than one.
- The greater stability of the chelate complex in comparison to normal complex is called chelate effect.

Example-

$Ni^{2+}(aq) + 6NH_3(aq) \rightleftharpoons [Ni(NH_3)_6]^{2+}(aq)$ $\quad K_{formation} = 10^8$

$Ni^{2+}(aq) + 3NH_2CH_2CH_2NH_2(aq) \rightleftharpoons [Ni(en)_3]^{2+}(aq)$ $\quad K_{formation} = 10^{18}$

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Quiz.

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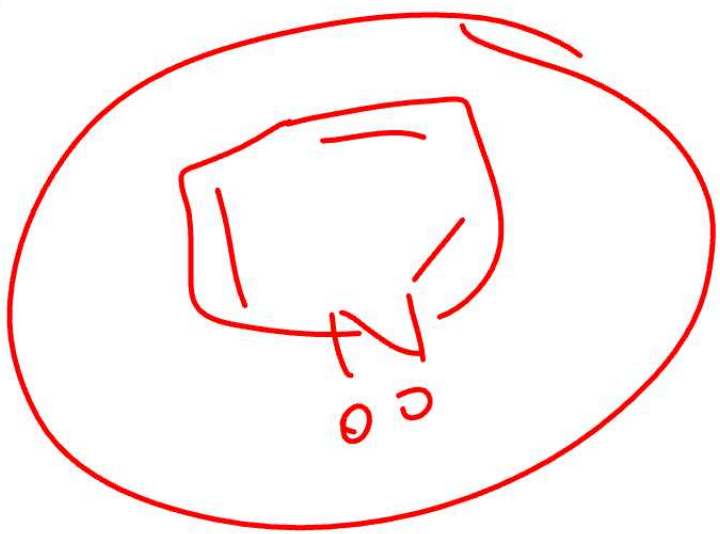
Which of the following ligands is not a chelating agent?

A. $EDTA^{4-}$

B. ethylenediamine

C. Oxalate

D. Pyridine



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Quiz.

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Which of the following ligands is not a chelating agent?

A $EDTA^{4-}$

B ethylenediamine

C Oxalate

D Pyridine

Solution

Pyridine is a monodentate ligand and thus can't act as a chelating agent.

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Quiz.

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The donor atoms in $EDTA^{4-}$ are

A Two N and Two O

B Two N and four O

C Four N and Two O

D Three N and three O

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Oxidation state of central metal ⟶

Sum of charges = net charge

$$\left[\underset{x}{Co}\left(\underset{0}{NH_3}\right)_6\right]^{+3}$$

$$x + 6(0) = +3$$

$$\boxed{x = +3}$$

O.S

$[Fe(CN)_3(NH_3)_3]$ ⟶ +3

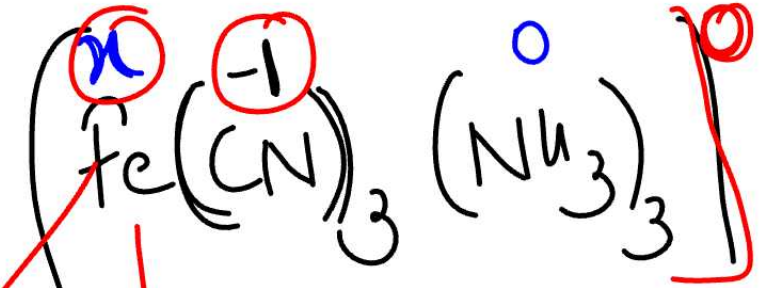
$[Ni(CO)_4]^0$ ⟶ 0

$[Ni(CN)_4]^{2-}$ ⟶ +2

$[NiCl_4]^{2-}$ ⟶ +2

$[Cu(NH_3)_4]^{+2}$ ⟶ +2

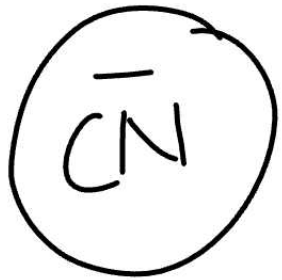
$[Cr(NH_3)_2(O)_2(H_2O)_2]^{-1}$ ⟶



+3

$x + (-1)3 + 0(3) = 0$

$x = +3$





$x + (-1)4 = -2$

$x = +2$

$$\left[x\,Cr(NH_3)_2\,(O)_2\,(H_2O)_2 \right]^{-1}$$

NH_3 → 0

$(O)_2$ → −2 (oxide)

H_2O → 0

$(O)_2$

(O_2)

$:\ddot{O}: + 2e^- \longrightarrow O^{2-}$

$x + 0(2) + (-2)2 + 0(2) = -1$

$x = +3$

(O_2)

JEE Main 2022 (Online) 29th July Evening Shift

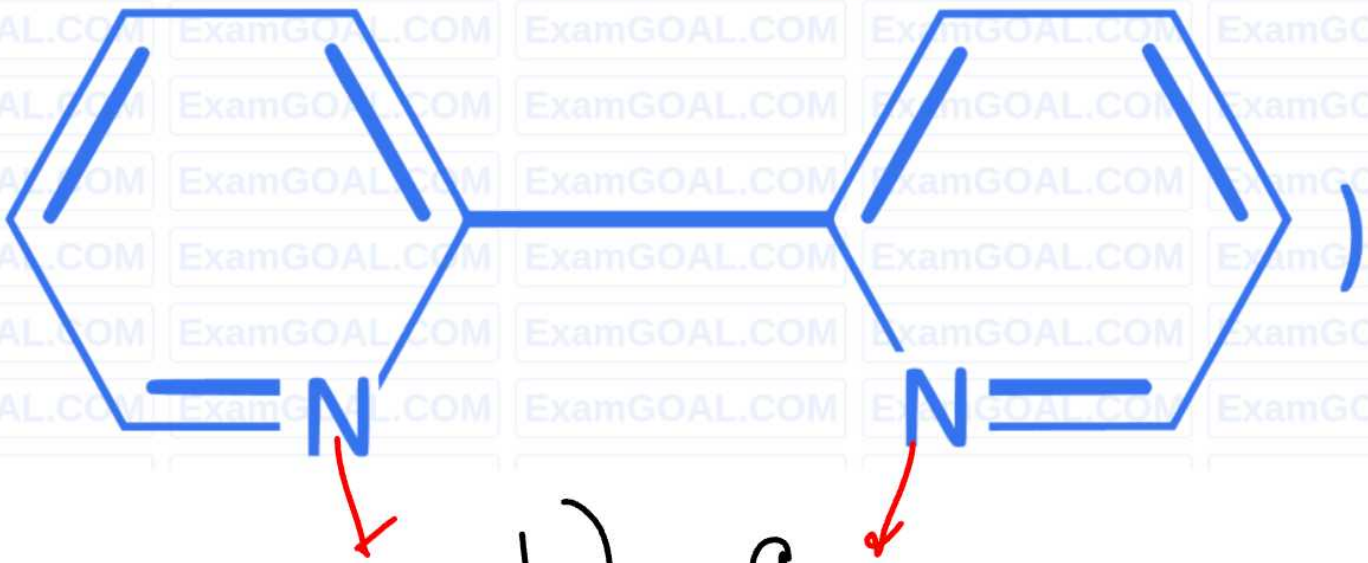
4 Numerical +4 -1

Sum of oxidation state (magnitude) and coordination number of cobalt in $Na[Co(bpy)Cl_4]$ is _____.

No. coordinate bonds

6 + 3 = 9

2 4

(Given : bpy = )

a) 8

b) 9

c) 6

d) 7

$$\overset{+1}{Na}\left[\overset{x}{Co}\overset{0}{(bpy)}\overset{-1}{Cl_4}\right]$$

$$+1 + x + 0 + (-1)4 = 0$$

$$\boxed{x = +3}$$

1 **JEE Main 2024 (Online) 9th April Morning Shift**

Numerical +4 -1

Number of ambidentate ligands among the following is _____.

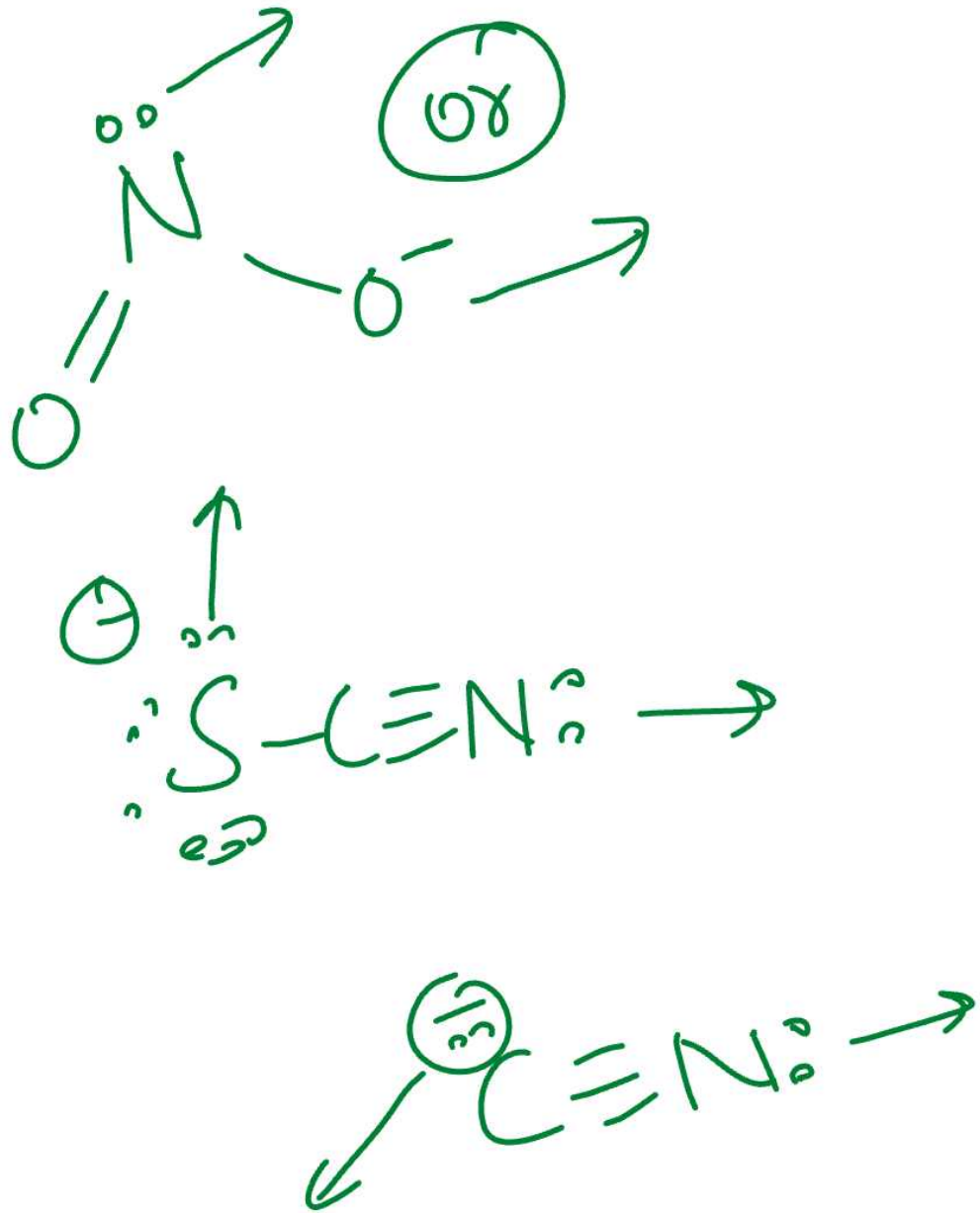
$NO_2^-, SCN^-, C_2O_4^{2-}, NH_3, CN^-, SO_4^{2-}, H_2O.$

a) 3

b) 4

c) 2

d) 5


or